Shankari V. Rajagopal

Harold C. Early Physics Early Career Professor, Assistant Professor of Physics
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Education

University of California, Santa Barbara
Ph.D., Physics — Jun 2019
Realizing and probing driven quantum systems with ultracold gases
M.A., Physics — May 2015

Massachusetts Institute of Technology
S.B., Physics — Jun 2012

Employment

University of Michigan
Harold C. Early Physics Early Career Professor — 2025-
Assistant Professor of Physics
- Interactive quantum dynamics with a Rydberg atom array in an optical cavity

Stanford University
Postdoctoral Research Scholar, Stanford Science Fellow — 2019-2024
Postdoctoral Adviser: Prof. Monika H. Schleier-Smith
- Studying spin models with Rydberg interactions
- Optimizing Rydberg interaction coherence to create metrologically useful entanglement with local interactions
- Simulating experimental realizations of Grover's algorithm for near-term quantum devices

University of California, Santa Barbara
Graduate Student Researcher — 2012-2019
Thesis Adviser: Prof. David M. Weld
Thesis: Realizing and probing driven quantum systems with ultracold gases
- Designed and built a lithium BEC machine
- Designed and built a strontium BEC machine
- Performed experimental quantum emulation of ultrafast ionization dynamics using ^{84}Sr BEC
- Measured electron-phason interactions in tunable cold-atom quasicrystal using ^{84}Sr BEC
- Studied Floquet-driven optical lattice systems using ^{7}Li BEC
- Designed a quantum gas microscope for strontium atoms

Massachusetts Institute of Technology
Undergraduate Researcher — 2009-2012

Fellowships & Awards

Stanford Science Fellowship, Stanford University	2020-2023
Chair's Outstanding Service Award, UCSB Physics	2016-2019
NSF Graduate Research Fellowship, NSF	2013-2018
2018 Mananya Tantiwiwat Fellowship Award, UCSB MLPS	2018
Outstanding TA Award, UCSB Physics	2016-2017

Publications

Fast and Robust Entanglement of Locally Interacting Spins for Quantum Sensing. N. U. Köylüoğlu, S.V. Rajagopal, G.L. Moreau, J.A. Hines, V. Khemani, and M. Schleier-Smith. *In preparation.*

Spin Squeezing by Rydberg Dressing in an Array of Atomic Ensembles. J.A. Hines, S.V. Rajagopal, G.L. Moreau, M.D. Wahrman, N.A. Lewis, O. Marković, and M. Schleier-Smith. Phys. Rev. Lett. **131**, 063401 (2023).

Number Partitioning with Grover's Algorithm in Central Spin Systems. G. Anikeeva, O. Marković, V. Borish, J.A. Hines, S.V. Rajagopal, E.S. Cooper, A. Periwal, A. Safavi-Naeini, E.J. Davis, and M. Schleier-Smith. PRX Quantum **2**, 020319 (2021).

Transverse-Field Ising Dynamics in a Rydberg-Dressed Atomic Gas. V. Borish, O. Marković, J.A. Hines, S.V. Rajagopal, and M. Schleier-Smith. Phys. Rev. Lett. **124**, 063601 (2020).

Phasonic spectroscopy of a tunable quantum quasicrystal. S.V. Rajagopal, T. Shimasaki, P.E. Dotti, M. Račiūnas, R. Senaratne, E. Anisimovas, A. Eckardt, and D.M. Weld. Phys. Rev. Lett. **123**, 223201 (2019).

Controlling and characterizing Floquet prethermalization in a driven system. K. Singh, K.M. Fujiwara, Z.A. Geiger, E.Q. Simmons, M. Lipatov, A. Cao, P.E. Dotti, S.V. Rajagopal, R. Senaratne, T. Shimasaki, M. Heyl, A. Eckardt, and D.M. Weld. Phys. Rev. X **9**, 041021 (2019).

Transport in Floquet-Bloch bands. K.M. Fujiwara, K. Singh, Z.A. Geiger, R. Senaratne, S.V. Rajagopal, M. Lipatov, and D.M. Weld. Phys. Rev. Lett. **122**, 010402 (2019).

Cold-Atom Quantum Simulation of Ultrafast Dynamics. R. Senaratne, S.V. Rajagopal, T. Shimasaki, P.E. Dotti, K.M. Fujiwara, K. Singh, Z.A. Geiger, and D.M. Weld. Nat. Comm. **9**, 2065 (2018).

Observation and uses of position-space Bloch oscillations in an ultracold gas. Z.A. Geiger, K.M. Fujiwara, K. Singh, R. Senaratne, S.V. Rajagopal, M. Lipatov, T. Shimasaki, R. Driben, V.V. Konotop, T. Meier, and D.M. Weld. Phys. Rev. Lett. **120**, 213201 (2018).

Experimental Realization of a Relativistic Harmonic Oscillator. K.M. Fujiwara, Z.A. Geiger, K. Singh, R. Senaratne, S.V. Rajagopal, M. Lipatov, T. Shimasaki, and D.M. Weld. New J. Phys. **20**, 063027 (2018).

Quantum Emulation of Extreme Non-equilibrium Phenomena with Trapped Atoms. S.V. Rajagopal, K.M. Fujiwara, R. Senaratne, K. Singh, Z.A. Geiger, and D.M. Weld. Ann. Phys. **529**, 1700008 (2017).

Effusive Atomic Oven Nozzle Design Using an Aligned Microcapillary Array. R. Senaratne, S.V. Rajagopal, Z. Geiger, K.M. Fujiwara, V. Lebedev, and D.M. Weld. Rev. Sci. Instrum. **86**, 023105 (2015).

Invited Talks	*Dynamical engineering of Rydberg atom systems for quantum-enhanced sensing, simulation, and optimization* University of Waterloo Quantum Innovators Workshop	2024
	Dynamical engineering of Rydberg atom systems for quantum-enhanced sensing and simulation Midwest Cold Atom Workshop (MCAW) Annual Meeting	2024
	Dynamical engineering of Rydberg atom systems for quantum-enhanced sensing and simulation Southwest Quantum Information and Technology (SQuInT) Annual Workshop	2024
	Dressed Rydberg atom systems for quantum-enhanced sensing Michigan Quantum Collaboratory Entanglement Meeting	2024
	Using neutral atoms for quantum metrology and simulation "What Physicists Do" Colloquium, Sonoma State University	2020
Contributed Talks	*Quantum metrology and optimization with Rydberg atoms* Joint Review Meeting, AFOSR Quantum Information Sciences	2023
	Spin squeezing in ensembles of neutral atoms via Rydberg dressing APS DAMOP	2023
	Spin squeezing with Rydberg-dressed interactions International Conference on Quantum Fluids and Solids	2021
	Spin squeezing with Rydberg-dressed interactions NSF CIQC/Quantum Foundry Joint Meeting, UC Santa Barbara	2021
	Quantum simulation and metrology with Rydberg-dressed Ising interactions ARO MURI Collaboration Meeting	2021
	Phasonic spectroscopy in a tunable quasicrystalline optical lattice APS DAMOP	2019
	Phasonic spectroscopy of tunable cold-atom quasicrystals APS March Meeting	2018
	Engineering quantum systems with light UCSB Photonics Society Light Science Workshop	2018
	Cold atom quantum emulation of ultrafast phenomena APS DAMOP	2017
	Quantum emulation with ultracold strontium and lithium Institute for Quantum Emulation meeting, UC Berkeley	2016
	Cold atom quantum emulation of ultrafast phenomena APS Annual Meeting of the Far West Section	2016

Teaching & Mentorship

Graduate Student Mentor 2019-present
Informal mentor — Six graduate students (junior and senior)
Rydberg Lab, Schleier-Smith Group, Stanford University

International Conference on Atomic Physics Mentor 2024-present
Formal mentor — Two graduate students

Summer Undergraduate Research Fellowship Mentor Jun-Oct 2021
Formal mentor — Undergrad research project, Stanford University

MAJOR Grant Summer Mentor Jun-Aug 2020
Formal mentor — Undergrad research project, Stanford University

Summer Institute for Mathematics and Science Aug 2017
Instructor — Center for Science & Engineering Partnerships, UCSB

UCSB Department of Physics Jan-Mar 2017
Teaching Assistant — Thermal and Statistical Physics, UCSB

Worster Fellowship Jun-Aug 2013
Mentor — Undergraduate research project, UCSB

Professional Service

Service to Journals
- Peer reviewer for *Physical Review Letters* 2024-present

Conference Organization
- 2024 Conferences for Undergraduate Women in Physics 2023-2024
  Member, Stanford Local Organizing Committee
  Stanford CUWiP
- 2019 Conferences for Undergraduate Women in Physics 2017-2019
  Co-Chair, UCSB Local Organizing Committee
  Member, National Organizing Committee
  UCSB CUWiP

Service to Universities
- Stanford Q-FARM Seminar Organizing Committee 2020-2023
  Organization of seminar series for Stanford quantum community
- Stanford Department of Physics, Equity and Inclusion Committee 2020-2022
  Focus on reading group organization and issues of reporting within the department. Member of APS-IDEA collaboration.
- Stanford Physics, Equity, and Identity (PIE) Program 2020-2021
  Panelist, *Building a More Equitable Physics Community*
- Stanford CAMPARE/Leadership Alliance 2020
  Committee to select students for program participation
- Climate Task Force, UCSB Department of Physics 2017-2018
  Focus on orientation improvement for graduate climate

Professional Service (con't)

Service to Academic Communities and the Public

- PUWMAS (Physics Undergraduate Women and Gender Minorities at Stanford) — 2019-2023
  Mentor for undergraduate and graduate students
- 2020 Conferences for Undergraduate Women in Physics — 2020
  Panelist, *Condensed Matter and AMO Physics Research*
  UC Irvine CUWiP
- UCSB Women in Physics — 2012-2019
  Treasurer, member
- 2018 Conferences for Undergraduate Women in Physics — 2018
  Panelist, *Building Community in Physics Departments*
  SoCal (Pomona/Harvey Mudd/Cal Poly) CUWiP
- UCSB Photonics Society — 2017-2018
  Speaker at local high schools for Women in Photonics Week
- UCSB Physics Circus — 2013-2016
  Ran physics demonstrations at elementary school fairs
- MIT PhysPOP (now MIT Discover Physics) — 2010-2012
  Co-founder/lead organizer, undergrad pre-orientation program
- MIT Society of Physics Students — 2009-2012
  Secretary, event organizer